Exercise: Extracting Data with a REST API

Explore how to extract data from third-party REST APIs using authenticated GET requests. Learn to process JSON and CSV responses, convert data into pandas DataFrames, and export it for ETL workflows.

Example

Let's see an example of using an API to extract data.

Suppose we're developers working for a cryptocurrency company. Our company is buying cryptocurrency data from a third-party data vendor that has an application that offers simple price reports for various currencies for the last 24 hours.

Let's take a quick tour through their app. First, click the live-app widget below to run the app. We should reach the home page of the application.

Next, click the tab "Symbols" on the top to reach a page showcasing all the cryptocurrencies available.

Your app can be found at: https://ed-5245074612158464-live-app.educative.run

Ask

Click to launch app!

This is all well and good, but how can we extract data from this third-party application? We can’t directly access their databases and servers. Luckily, the developers at the third-party company have made sure to create an API that allows authorized clients (such as developers at our company) to make GET requests and extract their data. They chose to authenticate their clients using an API key. Any request to their services with the correct URL and API key will be responded to with data.

For example, the script following the below live-app widget will send a GET request to the third-party app, along with the required URL and API key needed for authentication. Then, it will check if the response is successful or not and print a message accordingly. The API key for authenticating the requests is: `'12345'`.

The app's URL is in the top left corner of the widget. It has been automatically inserted into the script.

Ask

App URL

> **Note:** Make sure that the app has launched before running this script.

main.py Python 3.10.4

```
import requests
import pandas as pd
import datetime

# Define the URL
api_key = '12345'
url = f'{url}/api.symbols/api_key={api_key}'

# Send a GET requests to Crypto Currency App
response = requests.get(url)

# Check response code
if response.status_code == 200:
```

Ask

Run

Sending a GET request and authorizing the request with API key

Extracting a subset of the symbols

The developers at the third-party company have also made sure to add the functionality to extract a specific number of symbols instead of all of them. For example, if we wish to extract only the first three symbols, we can add the header `&&number_of_symbols=5` to the base URL to let the database at the third-party company know how much data we're interested in.

The following script will extract data from the third-party app, process it as JSON, convert it to a pandas DataFrame, and export it as CSV for later use.

> **Note:** Fill in the missing code in the following code widgets to complete the exercise. Control the amount of data to extract by adding the header `&&number_of_symbols=<<numberOfRows>>` to the URL.

main.py Python 3.10.4

```
import requests
import pandas as pd
import datetime

# Defining the URL
api_key = '12345'
url = f'{url}/api.symbols/api_key={api_key}<<InsertHere>>'

# Send a GET requests to Crypto Currency App
<<InsertHere>>

# Process the format as JSON
response_json = response.json()
```

Run

Ask

Send a GET request for a subset of the symbols

Extracting from a CSV format

The developers at the third-party company added another functionality to the API. We can now extract the data in formats other than JSON.

> **Note:** Fill in the missing code in the following code widgets to complete the exercise. To extract the data in CSV format, add `.csv` to the URL.

main.py Python 3.10.4

```
import requests
import pandas as pd
import datetime
import io

# Define the URL
api_key = '12345'
url = f'{url}/api.symbols/api_key={api_key}<<InsertHere>>.<<InsertHere>>'

# Send a GET requests to Crypto Currency App
<<InsertHere>>

# Convert the byte array into a string
```

Run

Send a GET request for data in CSV format

Let's explore the complete solutions for the tasks discussed above.

Solution code for extracting a subset of the symbols

Ask

main.py Python 3.10.4

```
import requests
import pandas as pd
```

```
import datetime

# Define the URL
api_key = '12345'
url = f'{url}/api.symbols/api_key={api_key}&&number_of_symbols=5'

# Send a GET requests to Crypto Currency App
response = requests.get(url)

# Process the format as JSON
```

Run

Send a GET request for a subset of the symbols

- **Line 10:** We send a GET request to the 3rd-party application's servers. We use the API key `'12345'` to authenticate. We use the header `&&number_of_symbols=5` to extract the first five symbols.
- **Line 13:** We process the response in JSON.
- **Line 16:** We convert to pandas DataFrame.
- **Line 20:** We export DataFrame to CSV.
- **Line 23:** We verify that the process is successful by loading the exported data.

Solution code for extracting from a CSV format

main.py Python 3.10.4

```
import requests
import pandas as pd
import datetime
import io

# Define the URL
api_key = '12345'
url = f'{url}/api.symbols/api_key={api_key}&&number_of_symbols=5.csv'
```

Ask

```
# Send a GET requests to Crypto Currency App
response = requests.get(url)
```

Run

Send a GET request for data in CSV format

- **Line 11:** We send a GET request to the third-party application's servers. We use the API key `'12345'` to authenticate. We use the header `&&number_of_symbols=5.csv` to extract the first five symbols in CSV format.
- **Line 14:** We decode the response using UTF-8 format.
- **Line 17:** We convert it into a regular file object we can read using the `io.StringIO()` method. Then, we read the file into pandas DataFrame.
- **Line 21:** We export DataFrame to CSV.
- **Line 24:** We verify that the process is successful by loading the exported data.

Conclusion

API is an excellent tool for regular users and developers to access data and communicate with other services. API establishes a common interface or a set of rules and protocols for requesting data or triggering specific actions. With APIs, we can safely and quickly interact with apps and websites, communicate with 3rd

Ask